TCS GenAI Lab — Verified Capabilities

Probed live 2026-07-26 against `https://genailab.tcs.in/v1`. Every row below was measured, not inferred from the catalog. `models.json` advertises **30** models; **7 are dead**. Do not trust `/v1/models` as an availability list.

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1 · What actually works

Model ID	Latency[1]	tok/s[2]	Max input[3]	JSON	Tools	Verdict
`genailab-maas-gpt-5.4`	1.3s	91	**≥900k** ✅ recall	✅	✅	**Workhorse. Whole-repo context.**
`genailab-maas-gpt-5.4-mini`	1.1s	**108**	250k–500k	✅	✅	**Fastest good model. Bulk codegen.**
`genailab-maas-gpt-5.4-nano`	1.4s	—	250k–500k	✅	✅	Cheap classify/extract
`genailab-maas-gpt-5.2`	2.1s	—	—	✅	✅	Works
`genailab-maas-gpt-5.1`	1.4s	77	—	✅	✅	Works
`genailab-maas-gpt-5.0`	2.0s	—	—	✅	✅	Reasoning tokens burned on trivial prompts
`azure/genailab-maas-gpt-5-mini`	1.6s	—	—	✅	✅	Works
`azure/genailab-maas-gpt-4.1`	1.1s	94	100k then **429**	✅	✅	⚠️ Rate-limits on big context
`azure/genailab-maas-gpt-4.1-mini`	1.2s	60	—	✅	✅	Works
`azure/genailab-maas-gpt-4.1-nano`	1.1s	—	—	✅	✅	Works
`genailab-maas-gpt-4o`	1.1s	—	128k	✅	✅	Slow at length (12.9s @100k)
`azure/genailab-maas-gpt-4o-mini`	1.1s	—	128k	✅	✅	Safe default, small jobs only
`genailab-maas-gpt-35-turbo`	1.2s	—	—	✅	✅	Legacy
`gemini-3-flash-preview`	1.4s	79	≥900k	✅	✅	⚠️ Erratic long-context recall
`gemini-3.1-pro-preview`	3.3s	100	≥900k	✅	✅	⚠️ Erratic long-context recall
`gemini-2.5-pro`	1.3s	124	≥500k	❌ raw	✅	⚠️ 1918 reasoning tok for 337 chars
`gemini-2.5-flash`	1.4s	—	—	✅	✅	Works
`gemini-2.5-flash-lite`	1.3s	—	—	❌ raw	✅	Works
`azure_ai/genailab-maas-DeepSeek-R1`	1.9s	—	—	❌	❌ 400	Leaks `<think>`; **no tool calling**
`azure/genailab-maas-text-embedding-3-large`	—	—	—	—	—	✅ **3072-dim**, working

[1] trivial prompt, warm. [2] measured on a real codegen task. [3] needle-in-haystack, verified recall.

2 · Dead — do not reference

Model ID	Failure
`genailab-maas-gpt-5.3-codex`	404 / 400 / 429 — never once succeeded
`genailab-maas-gpt-5.2-codex`	Succeeded once, then 404×2, 429×4 — **unusable**
`azure_ai/genailab-maas-Llama-3.2-90B-Vision-Instruct`	404 DeploymentNotFound
`azure_ai/genailab-maas-Llama-3.3-70B-Instruct`	404 DeploymentNotFound
`azure_ai/genailab-maas-Llama-4-Maverick-17B-128E-Instruct-FP8`	404 DeploymentNotFound
`azure_ai/genailab-maas-Phi-4-reasoning`	404 DeploymentNotFound
`genailab-maas-DeepSeek-V3-0324`	410 `model_deprecated`
`gemini-2.0-flash-001`	404 Publisher model not found

> **The two `codex` models are the trap.** Their names promise the best coding model on the gateway and they are the least reliable thing on it. Anyone who defaults `LLM_MODEL` to a codex ID loses the morning.

3 · Findings that change how you build

`gpt-5.4` **swallowed 900,000 tokens and recalled the needle exactly, in 19.5s.** That is the single most valuable fact here. It means an entire hackathon repo fits in one call — no chunking, no RAG, no vector DB for code-comprehension work.

Concurrency is free; total tokens are not. 16 simultaneous requests to the same model returned `16 ok / 0×429` for all five models tested, and an 8-model mixed fan-out completed 8/8 in **2.8s wall**. Every 429 observed came from *large-context* calls (250k+). The limiter is tokens-per-minute, not requests-per-minute:

- Small requests → parallelize hard (16-way is safe).
- 250k+ requests → serialize, one at a time.

Gemini long-context recall is unreliable. On a degenerate haystack, `gemini-3-flash` failed recall at 100k/250k/500k but passed at 900k; `gemini-3.1-pro` failed at every size. Real prose/code will do better than this adversarial filler, but do not put Gemini on a needle-retrieval path without testing your own data. The GPT-5.4 line recalled correctly at every size.

Two capability gaps: DeepSeek-R1 rejects `tools` with a 400 and leaks raw `<think>` blocks into `content`. Gemini `2.5-pro` / `2.5-flash-lite` won't emit bare JSON without fences even when instructed.

Correction to `models.md`: its "Known mismatch" note is resolved — `genailab-maas-gpt-4o` (no prefix) is correct and live. `azure/genailab-maas-gpt-4o` is not in the catalog and does not resolve. `CLAUDE.md` §6 in `hack2-master` lists that wrong ID plus `gemini-3-pro-preview`, which is also not a real ID (the live one is `gemini-3.1-pro-preview`).

4 · Reproduce

```
python probe.py   # availability, JSON, tool-calling across all 30
python probe3.py  # needle-in-haystack at 100k/250k/500k/900k
python probe4.py  # concurrency burst + mixed fan-out
```